



Re-politicizing the gender and climate change debate: The potential of feminist political ecology to engage with power *in action* in adaptation policies and projects in Nicaragua

Noémi Gonda*

Swedish University of Agricultural Sciences, Department of Urban and Rural Development, Uppsala, Sweden

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ABSTRACT

The time of gender-blind climate change policies and projects has passed. However, while research is increasingly moving away from understanding the relationship between gender and climate change in a linear, technocratic, and instrumental way, gender and climate change policy-makers and project practitioners are having difficulties operationalizing this progress. In the meantime, as climate change effects are increasingly felt worldwide, and because the policy context after the Paris Agreement (2015) is bringing new challenges for gender and equity concerns, (re-)politicizing the climate justice debate in a policy and project-relevant way is more crucial than ever. My aim in this article is to contribute to this endeavor by exploring how a feminist political ecology framework applied to a specific case study in Nicaragua—one of the countries most affected by climate change in the world—can generate new policy and project-relevant lessons and insights *from the ground* that can in turn strengthen the conceptual debate on gender and climate change adaptation. Based on ethnographic fieldwork carried out in 2013 and 2014, as well as eight years of professional experience as a development worker in Nicaragua, I discuss the *workings of power* in the feminist political ecology of climate change adaptation; in so doing I raise new questions that will, I hope, lead policy-makers and project practitioners to explore how adaptation processes could open up the conceptual possibility for emancipation, transformation, and new ways of living life in common.

1. Introduction

The effects of climate change are increasingly impacting women's and men's livelihoods across the globe. However, global warming, irregular rain patterns, and recurrent disasters are not the only reasons why the articulation of concerns for gender equality with adaptation to climate change is becoming extraordinarily challenging. Indeed, policy circumstances are evolving in ways that bode ill for both gender equality *and* adaptation, while progress in academic research on gender and climate change is insufficiently permeating the practical application of policy.¹

Illustrative of the type of decisions contemporary policy-makers are ready to make is the disappointing Paris Agreement² (United Nations,

2015). This first-ever universal and legally binding global climate deal does not reflect the significant progress that has recently been made in the scholarly understanding of the intersections between gender inequality and climate vulnerability. The problem is not just the fact that the Paris Agreement makes insufficient reference to gender. Rather, the Agreement's imperative of reaching a new, set, binding target in terms of limiting the rise in temperature is likely to direct the attention of policy-makers, project practitioners, and researchers toward further measuring the biophysical aspects of climate change to the detriment of focusing, instead, on people and questions of (social and gender) justice. Moreover, the increased importance that the Agreement gives to non-state interests and to market approaches by actors who often have little accountability to the public (Miller, 2007) is reason for concern

* Address: P.O. Box 7012, SE-750 07 Uppsala, Sweden.

E-mail address: noemi.gonda@slu.se.

¹ The intersection of gender inequality and climate vulnerability is of recent interest within policy spheres. It was only in 2001, at the seventh Conference of the Parties in Marrakech, that the need to include a gender perspective in the National Climate Change Adaptation Programs of Action (NAPA) was mentioned for the first time. Only nine years later, in 2010, at the sixteenth Conference of the Parties (COP) in Cancún, the necessity of designing climate change adaptation actions that take into account gender dimensions was emphasized (WEDO and GGCA, 2013).

² At the Paris climate conference (COP21) in December 2015, 195 countries adopted the first-ever universal, legally binding global climate deal. The agreement has set out a global action plan aimed at limiting global warming to well below 2 °C.

given the possible consequences that these factors are likely to have on the integration of gender and equity. Indeed, what is a *for profit* corporation's understanding of and interest in social justice, in general, and of (gender) equity in particular (Hale, 2016)? The Agreement's reinforced attention to adaptation instead of mitigation, while welcome, needs further scrutiny, as in many cases, responsibility for adaptation has to date been devolved to the local and individual levels, thereby assigning this duty to people who are not only ill-equipped to deal with climate change (Arora-Jonsson et al., 2016) but who have little responsibility for causing global warming. Similarly, the Agreement's focus on assigning responsibility (Lees, 2017), and thereby seeking direct causes for harm instead of analyzing wider structural constraints, is likely to bode ill for a people-centered approach and for gender equality.

Scholarship—and in particular feminist scholarship engaging with climate change (e.g. Tschakert et al., 2019, Nightingale, 2017, Arora-Jonsson et al., 2016, Eriksen et al., 2015, Kaijser and Kronsell, 2014, Carr and Thompson, 2014, Tuana, 2013, Tschakert, 2012, Slocum, 2004)—has a far more nuanced understanding of the complex interlinkages between gender inequality and climate vulnerability than that on display in the Paris Agreement. However, this understanding has permeated the policy and project spheres, and in general the practice, in very limited ways. In particular, feminist scholars have crucially contributed to a better understanding of how gender relations (and not only women) matter in climate change adaptation (Arora-Jonsson, 2011, MacGregor, 2010). Even more importantly, they have pushed to shift the attention from concerns about the biophysical effects of climate change toward the power-laden processes through which vulnerabilities are (re-)produced and possibly challenged (Tschakert and Tuana, 2013). Thus, they have helped to defy conceptualizations of adaptation as a mere technical fix and have argued instead for understanding it as a socio-political process (Eriksen et al., 2015) that is embedded in gender, caste, age, ethnicity, and other socially relevant hierarchies and in the workings of power in general. Key in the feminist approach is the intersectional lens that helps to address the practices through which social differences such as gender and ethnicity are enrolled in climate change adaptation. Intersectionality theory has even made it into the Intergovernmental Panel on Climate Change (IPCC)'s reports since 2014, in great part thanks to collaborating feminist IPCC authors, and ever more NGOs have been operationalizing this approach on the ground (see Oxfam Canada, 2014, Mangubhai and Capraro, 2015 on Christian Aid's experience, Gonda, 2014 on UNDP's experience).

Yet technical and institutional “solutions” to the effects of climate change still remain too attractive to project practitioners and donors. This is in part related to the fact that intersectionality is often misunderstood and/or misused in adaptation practice and theory; rather than serving to understand and challenge the workings of power that make people vulnerable to climate change, static conceptualizations of gender (and other social factors of difference) endure (e.g. Basu and Bazaz, 2018, Jobbins et al., 2016, Ensor and Harvey, 2015, Saeed et al., 2016) and become conceptual obstacles to seeing adaptation efforts as both productive of and limited by social inequalities. Thus, while in the last ten years or so the debate has been slowly but steadily moving away from a linear understanding of the relationship between gender inequality and climate vulnerability, when it comes to designing policies and actions to be implemented on the ground, especially in the Global South, policy-makers and practitioners have difficulties engaging with the processes of power that are critical in adaptation. For example, climate change adaptation projects in Nicaragua often recommend the use of water saving technologies or resistant seeds while they rarely engage in addressing the social and political causes (not just the consequences!) of water scarcity or of the disappearance of local resistant seed varieties. Thus, the climate change and gender debate seems to constantly drift away from its long-awaited “re-politicization” (Swyngedouw, 2014). The difficulties also stem from the fact that the socio-political processes of climate change adaptation are dynamic,

constantly evolving, and interlinked with many other socio-political processes; therefore, they are hard to grasp and engage with. Even more challenging for policy-makers and practitioners is figuring out how to concretely support the transformation of socio-political processes in such a way that not only they do not rely on and/or reproduce social and environmental injustices, but that they also support democratic environmental governance. Put somewhat differently: power is a difficult concept to grasp, let alone integrate into climate justice action. As a result, far too often, and sometimes despite much more ambitious aims, development institutions end up addressing only some of the “practical” potential negative effects of climate change (such as water or fuelwood scarcity) on existing gender inequality, as well as challenging some of the “concrete” ways in which the manifestations of gender inequality (for example the lack of participation of women in community or global actions) may limit people's capacities to adapt to climatic changes (Carr and Thompson, 2014). Hence, social, political and environmental transformations often remain an untackled topic.

By trying to understand the reasons for these difficulties and positioning the debate *on the ground*, this article shows how a detailed case study that is informed by key theoretical concepts derived from feminist political ecology differentiates itself from other gender and climate discussions and is important to broader policy debates around climate justice and democratic climate governance. In what follows, I first describe how ecofeminism and feminist environmentalism, two important gender and environment theories, have informed the approach to gender and climate change in Nicaragua, and why this has been problematic for questions of justice and democratic environmental governance in Nicaragua. In Section 3, I present feminist political ecology as a relevant framework for ensuring that the workings of power and social, political and environmental transformations remain on the agenda of climate change policy-makers and project practitioners. In Section 4, I break down the feminist political ecology lens into a fourfold conceptual framework inspired by the seminal work of Eriksen et al. (2015), Ahlborg and Nightingale (2018), as well as by recent efforts of feminist political ecologists engaging with climate change (Arora-Jonsson, 2011, Carr and Thompson, 2014, Cuomo, 2011, Kaijser and Kronsell, 2014, MacGregor, 2010, Singh, 2013, Tschakert and Tuana, 2013, Nightingale, 2017). I present four analytical foci (emotions, subjectivities, knowledges, and politics) that can help us engage with the power-laden processes that make rural women and men vulnerable to climate change, and that can be used by policy-makers and project practitioners to pose relevant questions regarding how their interventions may address climate justice. I illustrate the discussion based on the findings from ethnographic fieldwork that I carried out in rural Nicaragua between 2013 and 2014. In the final section, I tie together the four analytical foci and illustrate how they can be used to help re-politicizing the climate justice debate. I claim that this is particularly relevant in times of social, political, and environmental crisis, when, more than ever, there is the need to envision new democratic ways of living life in common.

2. The de-politicization of the gender and climate change debate in Nicaragua

Two important gender and environment theories—ecofeminism and feminist environmentalism—have informed policy and project approaches to gender and climate change in Nicaragua since 2010. In this section, I explain why the way they have been used by policy-makers and project practitioners has contributed to the de-politicization of the gender and climate change debate.

While theoretical academic work on gender and climate change is hard to compare with knowledge that is produced through projects and programs around gender and climate change, it is important to understand how policies and projects draw upon certain theories and with what effects. In the Nicaraguan case, ecofeminism and feminist environmentalism have permeated climate change policies and projects

carrying with them the assumptions that gender roles are immutable and that the numeric participation of women in policies and projects is equal to having a feminist perspective. In the meantime, they have not been envisaging the radical transformation of the societal system that constantly reproduces unequal gender relations and the causes of climate change. Later in this article, I will argue that these shortcomings that have ultimately contributed to de-politicizing the climate change debate and to instrumentalizing women and other already marginalized population groups have, in part, contributed to the 2018 political crisis in Nicaragua.

Nicaragua is considered the fourth most climate-change-affected country in the world for the 1997–2016 period (Eckstein et al., 2017). For this reason, addressing climate change by supporting rural populations to adapt to the present and foreseen effects of climate change has become one of the country's environmental priorities since the early 2010s. Indeed, since that time, the Nicaraguan government has identified climate change adaptation as one of its key priorities (Campos Cubas et al., 2012), as can be seen by comparing the importance given to climate change in the 2012 National Human Development Plan (GoN, 2012) to its importance in the preceding 2003 National Development Plan (GoN, 2003).

Since the 2010 Nicaraguan climate change adaptation strategy, women have been presented as saviors in the face of climate change, which reflects a conceptualization of the relationship between gender and the environment that can be found in ecofeminist writings. Ecofeminism gained peak popularity among NGOs and feminist activists in the 1980s. As a theory that encompasses both gender and the environment, it includes the following tenets: First, it suggests that there are connections between the domination and oppression of women and the domination and oppression of nature. Second, it purports that within patriarchal thinking, women are identified with nature while men are identified with culture, and that the latter is superior to the former. Third, it claims that women are particularly suited to ending the domination of nature because they are closer to nature than men. Finally, it assumes that feminist and environmentalist movements have a common goal of achieving less hierarchy and more equality and are, hence, defending common interests (Agarwal, 1992).

The ecofeminist influence in the Nicaraguan climate change policy narrative helped create an idea of (in reality false) inclusiveness. Indeed, poor rural women's (supposed) concerns were brought to the fore but only at the rhetorical level. This happened within the context of the Nicaraguan Sandinista government coming to power in 2007 and having a narrative on the environment that denounced the destructive character of neoliberalism and called for instantiating a new era characterized by humans living in harmony with "Mother Earth" (GoN, 2012, 2010). This discourse feminized nature by attributing to it traits such as vulnerability and respectability that are recognized as "feminine." References to these traits were then used to justify the argument that the earth has to be taken care of and respected—as if it were our own mother. Another important feature of the Nicaraguan climate change policy discourse was that it naturalized women's capacities to engage in climate change adaptation. Indeed, the narrative and the related actions encouraged women's participation in environmental management and prioritized their contributions to environmental protection over the actions of men. The government justified these policies of prioritizing women with the argument that women were not only the ones most affected by climate change but that they also had a "special" and "natural" connection to nature and were, therefore, especially prone to fighting the negative effects of climate change and of engaging in adaptive actions.

Ecofeminism and the development approaches that it inspired—the Women In Development (WID) and, later, the Women, Environment, and Development (WED) approaches—call for a greater consideration of women in development as well as in environmental management policy and practice (Reeves and Baden, 2000) by directing the attention toward the existing gendered division of labor and (static) gender roles.

These approaches call for more attention to the ways in which women's contributions might help alleviate the effects of climate change by focusing on what are "traditionally" constructed as women's tasks such as gathering water and fuelwood and home-garden production. Razavi and Miller (1995) attribute the rapid popularity achieved by the WID and WED approaches among development and environmental agencies during the 1980s and 1990s to the fact that they articulated a triple-win relation among social justice, economic efficiency, and sustainable environmental management.

However, this apparently winning trifecta has been revealed to be often counter-productive (e.g. Escobar, 1995, Reeves and Baden, 2000). In particular, the WID and WED approaches operationalized by projects and policies have often contributed to increasing women's workloads without advancing their positions in society or allowing them to exert more power in decision-making within their families or communities (Escobar, 1995). In addition, these approaches often conflated notions of gender with actual women and essentialized women as a homogeneous group, in addition to failing to raise strategic issues for women by prioritizing their practical needs (Leach, 2007).³ Yet the integration of ecofeminist views into development policies was, in a certain way, useful: before that, women were mostly invisible to development strategists (Sturgeon, 1999). Hence, the main achievement of ecofeminist theories as well as the WID and the WED approaches has been to direct attention to the fact that, if development was to be achieved, attention had to be accorded to women.

The problematic visibility of women and the construction of them as the saviors in the face of climate change within the Nicaraguan climate change policy narrative have been intersecting with another approach that views women as the main victims of climate change. To some extent, this victimizing narrative can also be found in the academic literature and in some development organizations' publications that purport to be working on climate change with a gender perspective (see for example Cannon and Müller-Mahn, 2010 for the academic literature, Aguilar et al., 2015 for IUCN, Habtezion, 2013 for UNDP).

This narrative that victimizes women is as problematic as the narrative that views women as saviors in the face of climate change. Its conceptualization of the relationship between gender and the environment relates to the theoretical underpinnings of feminist environmentalism, which was translated into development approaches in the 1990s and 2000s under the name of Gender and Development (GAD) and Gender, Environment, and Development (GED). Feminist environmentalism seeks to address the material realities in which women of different classes, ages, and ethnicities are rooted and which might affect their responses to environmental degradation (Agarwal, 1992). It assumes that people's material conditions, such as whether they have access to land, information, communication means, or training, influence their relationship to the environment (Nightingale, 2006). A feminist environmentalist perspective translates into activities aimed at challenging the structural causes of environmental degradation and gender inequality. Challenging these causes constitutes precisely its strength.

Nevertheless, because feminist environmentalism focuses mainly on the material relations between gender and the environment, when this theoretical perspective is applied, it is often technical and local solutions to climate change that are put forward. Such solutions seek, for example, to facilitate access to water or fuelwood. Sometimes they aim

³ Scholars such as Niamh Moore (2008) suggest a more complicated account of ecofeminism that would not see essentialism as an inherent feature. While I agree with the more complicated account, and welcome recent efforts that use ecofeminism to discuss how to include gender in alternative development approaches such as degrowth (Prieto Pérez and Domínguez-Serrano, 2017), my point here is to show how Nicaraguan climate change policy has used ecofeminism, which I argue has been done in an essentializing and problematic manner.

to help increase women's income, because vulnerability is often assimilated to poverty and to women. Since material relations with the environment are negotiated “on the ground”, an important inconvenience of feminist environmentalism is that it insufficiently addresses the multi-scalar interactions between gender and climate change beyond the material.

Attention to the workings of power and to the possibility of social, political and environmental transformations have been sadly missing in the way ecofeminism and feminist environmentalism have been taken up by Nicaraguan climate change policies and projects. The understanding of the connections between gender and climate change have been mostly reduced to the claim that women are best positioned to fight climate change, and are the most affected by it, and to measures that merely reinforce “traditional” gender roles. To re-politicize the debate, engage with the workings of power and touch upon questions of transformation, justice, and democratic climate governance, a much more complex understanding of the connections between gender and climate change is needed.

3. The potential of feminist political ecology to apprehend the workings of power in climate change adaptation

In this section, I discuss key theoretical concepts of feminist political ecology to show how this approach differentiates itself from other gender and climate approaches and how it can contribute to re-politicizing the climate change debate, especially *on the ground*.

3.1. Feminist political ecology

For feminist political ecology, the relationship between gender and the environment is conceived of as a dynamic process in which gender and the environment play important roles but culture and society are also of core significance (Nightingale, 2006). Each of these elements (gender, the environment, culture, and society) and the relations between them are embedded in, shaped by, and in turn construct power relations. Since Rocheleau and her colleagues' landmark work (1996), feminist political ecology has evolved considerably. The early feminist political ecology focused on three themes: “gendered environmental knowledges, gendered environmental rights and responsibilities, and gendered environmental politics and grassroots activism” (Elmhirst, 2011, 129). Since then, three important factors have modified the field of feminist political ecology. Poststructuralist and performative approaches in feminist theory have concentrated feminist scholars' attention on the transformation of gendered power relations, rather than on static gender roles. They have also drawn attention to the effects of (gendered) discourses (Elmhirst, 2011) that have the potential to create their (gendered) citizen-subjects and to the performance of subjectivities (Nightingale, 2013). Perhaps most importantly, this evolution in the scholarship opened up the space to talk about emancipation and transformation in the context of, and in response to, environmental changes (Djoudi et al., 2016, Manuel-Navarrete and Pelling, 2015, Eriksen et al. 2015, Manuel-Navarrete, 2010).

Also, recent work in feminist political ecology takes into account the effects of new forms of environmental governance intervention linked to specific political contexts and to development cooperation (Elmhirst, 2011) as well as the effects of populations' increased mobility (Nightingale, 2006) and of the fact that ideas travel increasingly easily in the globalized world across scales (Tschakert, 2012). Importantly, the new feminist political ecology is intersectional, moving beyond a mere focus on gender, as people are not “just” men or women: a variety of social factors such as ethnicity, age, and class intersect to shape both oppressions and privileges that ultimately add up in the processes that make particular people vulnerable to social and environmental changes (Kaijser and Kronsell, 2014, Nightingale, 2011b).

New forms of intervention and governance arrangements, as well as the fact that people and ideas are increasingly mobile, generate new,

sometimes hybrid discourses on the environment. For example, the Nicaraguan gendered climate change narrative has been intersecting with more mainstream narratives that promote market-based solutions to environmental degradation—such as payment for ecological services—or that see women's participation in environmental management and poverty reduction as “smart economics,” to use the words of former World Bank director Robert Zoellick (2011).

3.2. Feminist political ecology of climate change

My intention in this article is to push the feminist political ecology framework further than some feminist political ecologists typically do. Indeed, climate change as a research topic has been challenging for many feminist political ecologists especially when it comes to producing ethnographic studies that succeed in bridging scales and that can, at the same time, provide ideas for designing just climate policies and interventions. But the scalar politics (Swyngedouw, 2004) of knowledge creation are particularly key for feminist scholars who want to engage with climate change adaptation. Engaging with scalar politics is primordial because they are often the means through which social inequalities are (re)produced (Eriksen et al., 2015). Another aspect that has been challenging for feminist political ecologists is the engagement with policy-relevant and practitioner debates on gender and climate change (Westholm and Arora-Jonsson, 2015, and I have myself attempted to do so in Gonda, 2016): feminist political ecology has not translated into development approaches in the same relatively straightforward ways in which ecofeminism and feminist environmentalism translated into WID and GAD respectively.

Working on climate change across scales through a feminist political ecology lens means connecting what happens at the emotional and embodied levels in the context of climate change with global climate change politics and the politics of knowledge-making on climate change. On the one hand, it is necessary to engage with the geoscientists of the climate change community who tend to generate research that is still too often technocratic (Löfbrand et al., 2015, Nightingale, 2017, Ojha et al., 2016) and top-down, often insufficiently usable for on-the-ground interventions, but very much influenced by the interests of particular lobbies. Feminist political ecology's pertinence lies precisely in its potential to give an account of the scalar politics of climate change and its interest in discourses, power relations, and the workings of power, justice, hegemonies, and development (Tschakert, 2012, Harcourt and Nelson, 2015). On the other hand, feminist political ecology's potential to draw attention to the individual level brings another important and under-scrutinized topic into the debate: that of the embodied and emotional aspects of climate vulnerability (González-Hidalgo and Zografos, 2017, Nightingale, 2013, Singh, 2013, Sultana, 2011, 2015).

In sum, a feminist political ecology perspective on climate change can foster (re-) politicized approaches to climate change as well as challenge impact-oriented and technocratic policies and interventions while revealing political influences and/or the influence of particular lobbies. It brings in a new perspective on how to address gender inequality, gendered unequal power relations, and gendered social injustices that are embedded in both the policy and practice of climate change adaptation (Chandra et al., 2017). But how to engage with the workings of power in the inter-scalar processes that make particular people vulnerable to climate change? The next paragraphs present the analytical framework that I will apply to my Nicaraguan case study in Section 4 to do so.

3.3. Vulnerability and the workings of power

The central focus of a feminist political ecology perspective on climate change should be the workings of power (Ahlborg and Nightingale, 2018) in the processes that make people vulnerable. The workings of power cannot be apprehended solely through discussions

that are limited to understanding who is empowered and who is not; just like understanding vulnerability is not about assessing the levels of vulnerability of population groups. Rather, understanding vulnerability is about apprehending the very (interconnected, dynamic, ever-changing) processes that make specific people vulnerable to climate change. These processes are intersectional (Kaijser and Kronsell, 2014, Nightingale, 2011b, Tschakert, 2012): social difference (gender, ethnicity, age, etc.) emerges and is enrolled in the everyday practice (Nightingale, 2011b) of climate change adaptation. Thus, engaging with the processes that make particular women and men vulnerable to climate change means working with the power structures that intersectionality creates—for example, when marital status (such as being single) intersects with gender (such as being a woman) in specific empowering and disempowering ways that can occur simultaneously. For example, single mothers are recognized as an especially vulnerable group in the discourse of NGOs working on climate change adaptation. While being a single mother may help some women access project support more easily than others, the same discourse that constructs them as vulnerable can also diminish their agency. Single men's specific vulnerabilities in the face of increasing water and wood scarcity (in the case of widowers or men with limited mobility) may not be recognized precisely because of this biased focus on women and on roles that are “traditionally” attributed to women. How to engage with these interlinked, ever-changing processes that create vulnerability? This is the challenge that a feminist political ecology perspective can help us undertake.

In general, power is a difficult concept to grasp, and project practitioners and policy-makers have a hard time envisioning measures that would help challenge unequal power relations, let alone work towards socio-environmental transformations whose outcomes may be uncertain. To engage with the workings of power in the processes that make particular people vulnerable to climate change, I argue that what is required is a fourfold analytical focus that bridges scales—from the emotional and embodied levels to the level of global politics. My analytical focus is inspired by the seminal work of Eriksen et al. (2015) as well as Ahlborg and Nightingale (2018), which I find to be particularly pertinent. It also builds on the work of feminist political ecologists who are engaging with climate change (Arora-Jonsson, 2011, Carr and Thompson, 2014, Cuomo, 2011, Kaijser and Kronsell, 2014, MacGregor, 2010, Tschakert and Tuana, 2013, Nightingale, 2017) and who themselves mobilize a distinctly Foucauldian and feminist understanding of power (Foucault, 1982, Allen, 2002, Butler, 1997) by pushing the study of power toward the study of power in action rather than discussing its inherent nature or who holds power over whom.

3.4. Methods

My analysis is based on four years of doctoral research (2012–2016) on the politics of gender and climate change in Nicaragua. While the discussion in this article focuses mostly on policy discourse⁴ and the observation of climate change project activities in 2013 and 2014 in two rural communities,⁵ it is also enriched by my experience as a former development NGO worker in Nicaragua (2004–2010) and by 108 qualitative interviews I conducted with rural women and men, people working in organizations, and with researchers and activists. The ethnographic examples I use in the Discussion Section are from

⁴ To understand the policy approach, during 2013 and 2014 I attended 38 different national and regional events on development and climate change in Nicaragua such as seminars, workshops, and *fora*.

⁵ The community of El Nancite in the municipality of Telpaneca, department of Madriz, situated in the “Dry Corridor” of Nicaragua, and the community of El Pijibay in the more humid municipality of El Rama on the former agrarian frontier, in the Autonomous Region of the South Caribbean Coast of Nicaragua (the names of the communities and the people mentioned in this article have been changed for anonymity reasons).

2013 and 2014.

In the following sections, I analyze in detail how the combination of this fourfold focus on (1) the emotional experiences of adaptation, (2) the subjectivities that climate change adaptation processes create or challenge, (3) climate change adaptation knowledges, and (4) climate change adaptation politics can help policy-makers and project practitioners apprehend the *workings of power* in the processes that make rural women and men vulnerable to climate change.

4. Discussion: Engaging with power from the emotional experiences of adaptation to global climate change politics

In this section, I use the Nicaraguan example as a way of understanding the *workings of power* in the practice of adaptation and ultimately as a way of thinking about socio-environmental transformations in policy and project-relevant ways.

4.1. Emotions

Emotions⁶ matter in the climate change debate because they bring in a relational view of vulnerability, as the power to affect and to be affected (Tschakert and Tuana, 2013, Anderson, 2005, Butler, 2009). If not manipulated by discourses, policies, or interventions, emotions can help us envision new conditions for living life in common in a sustainable, equitable, and just way, even in the context of climate change. Emotions are “embodied experiences that imply individual and collective ways of thinking, engagements, expressions and relationships among humans and their environments, which subsequently inform political ecology relationships among people, their environments and conflicts” (González-Hidalgo and Zografos, 2017, 63). Thus, emotions can reveal “how people's sense of self is shaped by their affective capacity to respond to other bodies, both human and non-human” (Singh, 2013, 190).

Researchers, policy-makers, and project practitioners need to be aware of the potential for political manipulation of the emotions (Sharp, 2009), a particularly understudied topic in political ecology, human geography, and development studies. Such possibility for manipulation is, however, exceptionally relevant for climate change as climate change tends to be presented as the ultimate catastrophe of our time: therefore, it is appealing to our emotions. Without questioning the latter statement, I argue that it is imperative to ask how invoking emotional responses in climate change narratives, policies, and interventions in order to make people take action can end up helping to maintain the status quo in the climate change debate. For example, in Nicaragua, climate change policies and projects have been encouraging women's affective labor of taking care of the environment. This helped the government to construct the subjects of Nicaraguan politics—the flagship subjects of these politics being the compliant, subordinated, grateful women who are close to “Mother Earth” and who were made the main beneficiaries of climate change adaptation projects. In general, climate change narratives often put many environmental responsibilities onto certain individuals (Cuomo, 2011)—something that can be emotionally overwhelming, especially when questions of broader accountability and of the possibility of radical socio-environmental transformation are not raised.

Beyond their ability to be manipulated, emotions are the openings through which new ways of living life in common will emerge (Singh,

⁶ While, I am aware of the debate between affective and emotional geographies that is centered around questions of representability—emotional geographies emphasize the significance of *expressed* emotions while affective geographies focus on quality of life *beyond* cognition that is always interpersonal but *inexpressible* (Pile, 2010)—here I use some important commonalities of emotional and affective theories to bring in the importance of looking at emotions in the climate change debate.

2013); put somewhat differently, they may open the pathways toward transformation. To produce new, just, sustainable, and equitable ways of living life in common in the context of climate change, there is a need to build on positive emotions that have not been manipulated. Scholars, policy-makers, and project practitioners interested in climate justice have a particular responsibility to contribute creatively to the process of building new emotional relations across scales in the context of climate change: they open up the possibility for new conceptual understandings centered around notions of care, affect, and living life in common.

4.2. Subjectivities

The subjectivities that are created and/or challenged during adaptation processes are another key analytical focus to engage with. For Butler, subjectivation is broadly understood as the “making of the subject” (1997). It is “a kind of power that not only unilaterally acts on a given individual as a form of domination, but also *activates* or forms the subject” (Butler, 1997, 87 italics in original). Subjectivation occurs, for example, through the discourse of climate change adaptation politics when it creates the figures of the environmentalist woman or the culprit smallholder farmer. Some subjects comply with the discourse and try to become what they are “supposed” to be, while others resist or strategically use the discourse to become “something” else, sometimes outside the discourse. In this process, a particularly interesting aspect relates to self-conceptualizations (of one’s own vulnerability in the face of climate change, for example) as they can be analyzed as strategies of “governing the self” (Foucault, 1982) that become also part of the everyday enactments of climate change adaptation. Hence, subjectivities that emerge through “the ways in which people are brought into relations of power” (Nightingale, 2011a, 123) are always “in the making” (Sundberg, 2004): to study them, one needs to be “attentive to how disciplining discourses and practices are invoked, enacted, (re)configured, subverted, and transformed by individuals” (Sundberg, 2004, 46–47).

Understanding the effects of subjectivation processes and how they can be subverted is key. For example, the effects of the narrative that victimizes women can be illustrated with reference to events that occurred at a workshop on gender that I attended along with approximately thirty participants from four communities of Nicaragua’s “Dry Corridor.” The event was supposed to help collect information on gender roles and relationships for a baseline study of a UNDP climate change adaptation project (Workshop in the community of Las Lajitas, 08/04/2014). I observed that the workshop’s exercises were designed in such a way that the participants would confirm and conform to the female stereotype of the victim in climate change. The facilitators who were hired to conduct the event, on the basis of which they had to elaborate a report, asked the group of participants to identify what was happening in the majority of situations. The questions that were asked included whether women or men were in charge of fetching water and wood; who got up earlier and who went to bed latest; or who was lacking ownership and control over the means of production, called by the facilitators “adaptation technologies.” The questions suggested that women were expected to answer that they were the ones who fetched firewood and water, that they were the ones who got up earlier as compared to men, and that they did not have control over the means of agricultural production. Very few of the answers given by participants matched the expectations of the facilitators. For example, a sixty-year-old single woman from El Nancite, who has never been married, who is a mother of an adult daughter, and who lives and works alone on her ten-hectare farm, gave the following unexpected answer:

Facilitator (addressing all women): Who [among you] helps her husband in agricultural work?

Doña Leonor: I am the one who works [personally] on the land.
(Paraphrase from fieldnotes 08/04/2014)

At the workshop, as the debate dragged on producing other similar

“unexpected” answers, and the facilitators of the workshop wanted to move on but not without reaching a consensus that would appear in their report, they started suggesting percentages. For example, they recommended that the participants agree to the fact that in 30 percent of cases, it is men who fetch water and in 70 percent of cases, it is women. They also suggested that participants agree on the fact that women sleep five to six hours per night, while men sleep seven to eight hours (fieldnotes 08/08/2014). This approach did not help the facilitators understand either the reasons why some situations were particular (such as the case of Doña Leonor), or the factors that are at the origin of some gendered vulnerabilities. The timid efforts to break out from the essentializing picture of women without agency (example of Doña Leonor) were stopped by the facilitators, thus purposefully or inadvertently reinforcing the stereotyped vulnerable female subject in the face of climate change, which, I argue, can ultimately legitimize household inequity.

I claim that this vignette from the field is illustrative of the sort of Foucauldian device enabled by interventions to discursively create the feminized, vulnerable subjects of climate change policies and projects. The narrative of women’s vulnerability was reinforced by the very processes through which (1) knowledge for the baseline study was generated and (2) that same knowledge would ultimately be presented as not only having been gathered through a bottom-up participatory process, but also that it was scientific and representative precisely because it included percentages. It is interesting to highlight that while development and climate change projects are increasingly designed for women (and try to include women in their execution), they still contribute to maintaining male dominance (Sundberg, 2004), just as when they were eminently designed for men.

Additionally, the unequal power relations between the facilitators and the participants (re-)enforced the making of feminine vulnerabilities. These power relations, stemming from the differing social status between the facilitators (who were two non-indigenous, educated, urban Nicaraguan men from outside the communities and who introduced themselves as “engineers”), and the attendees (mostly uneducated rural women and men), were strikingly apparent in the way the facilitators enforced the knowledge that they considered to be valid; the power differential was also evident in the extreme poverty in which some community inhabitants lived, which made them welcome any project activity that provided them with food, especially during the period in which there was no off-farm work and nothing to harvest (usually between February and May, precisely when projects like to organize their activities because community inhabitants, at least those who do not migrate, are available). Eventually, among the women who attended the workshop, the majority (including Doña Leonor) acquiesced to the discursive constructions of their vulnerability, and they did not express major discontent with the outcome of the workshop.

Given this, the first challenge is to better understand the processes of subjugation: this means to challenge the practices through which it is reproduced, including the performance of subjugated subjectivities. The second and related challenge is to support the emergence of emancipatory subjectivities. In the community of El Nancite, there was a women’s group formed in 2003 that was called *Las Vulnerables*⁷ but whose members decided in 2014 to change their name to Grupo San José. Its members told me that the group was formed to undertake a health project that included attending to small children in the community. One of the founders of the group, Doña Rosa, a twenty-eight-year-old married woman and mother of two girls, explained to me that a male technician suggested that they should adopt the name *Las Vulnerables*:

He told us that it was a good name for the group because some of us were single mothers ... and ... the others ... well, the others ... we were women.

⁷ The Vulnerables.

(Interview with Doña Rosa, El Nancite, 11/04/2014).

While essentially symbolic, the name change is also potentially emancipatory. When the subject knows and /or governs herself as she is supposed to, according to the authority (represented here by the project), by enacting her subjectivity as she is supposed to, she reinforces this authority. But when subjectivities shift, and when the speech and/or actions related to these subjectivities transform, they have the potential to alter the subjectivation process. Emancipatory subjectivities are indeed key for socio-environmental transformation, they are “casting individuals with a choice to consent, evade, or transgress established authority in order to reduce both inequalities and (...) (environmental) risks” (Manuel-Navarrete and Pelling, 2015, in Tschakert et al., 2016, 184).

In sum, a focus on subjectivities in the climate change adaptation debate shows the actual power that field encounters have and the potential damage caused by the forms of subjectification described above. It also opens up the possibility of talking about the role of policies and projects in supporting beneficiaries’ and participants’ emancipation. It raises the question of what emancipatory subjectivities would look like in the context of a changing environment and society, and how they would be linked with positive emotions and relations of care.

4.3. Climate change adaptation knowledges

Knowledge politics are the third important analytical focus to engage with in the processes that make particular people vulnerable to climate change. Feminist theory is especially suitable to disclosing power relations in environmental knowledge politics (Code, 2008, Haraway, 1988, Nightingale, 2003, Slocum, 2004) by showing that “what we know and do not know [in the issue of climate change] is inextricably interlinked with issues of justice” (Tuana, 2013, 14). For example, in Nicaragua there have been intentional efforts to discursively render invisible the important role plantations play in causing deforestation, and these efforts have run parallel to those blaming smallholders for cutting down forests and thereby contributing to climate change. This particular framing of the “issue” of deforestation—and in particular, when used by policy makers and projects to justify the necessity of planting trees or implementing agro-forestry systems as adaptive actions—is deeply enmeshed with ethnicity, social inequality, and sometimes gender-related privileges. The question of positionality or situatedness in relation to climate change knowledges is crucial (Ahlborg and Nightingale, 2018, Cote and Nightingale, 2012, Nightingale, 2003, Tschakert and Tuana, 2013): Who defines what the “problem” is and who is responsible for causing and solving it? How is knowledge-production taking place based on these definitions? How are “valuable” and “invaluable” knowledges constructed, and what are the channels through which such knowledges are “supposed” to be respectively transmitted or dismissed?

The type of climate change studies that are commissioned and the way they get translated into policies and actions can reveal the tenets of knowledge politics. Shifting to cocoa—a product of agro-forestry—was often presented in Nicaraguan climate change policies and projects as the “miraculous” solution to cope with the effects of climate change (Mora Flores, 2014). I argue that this prioritization of cocoa production in adaptation policies and projects in Nicaragua was in great part a response to the economic interests of politically and/or economically powerful actors (Gonda, 2017). These actors were among the wealthiest members of the coffee lobby: politically influential coffee producers whose farming systems were increasingly in crisis due to climate change effects and who were interested in shifting their production from coffee to cocoa. Actually, some of them were at the negotiation table when the Nicaraguan sectorial climate adaptation strategies (GoN,

2013) were discussed together with ministers and researchers, as one of my interviewees recounted. The presence of the coffee lobby has greatly influenced the fact that impact studies on the coffee sector had a significant impact on Nicaraguan climate change adaptation policies, while other studies (for example on maize and beans production, a sector that relates to smallholders’ livelihoods) have been mostly taken up by NGOs (paraphrasing from an interview with a Nicaraguan climate change researcher, Managua 28/10/2014).

Another important question is how local knowledge is brought into the debate, who is asked to participate in knowledge production at the local level, and how creatively unequal power relations are used to ensure participation of people with different skills, backgrounds, and knowledge. For example, in Nicaragua in 2013, in the context of a series of workshops called Communitarian Diploma on Climate Change, organized by two international NGOs and a university, considerable efforts were made to promote rural women’s participation in trainings on climate change. The project provided money to the families that sent a female, instead of a male, member to attend the training. While this was probably well-intended, some women who otherwise were not interested in attending the training were forced to do so because of the economic incentive. Additionally, women risked being taken advantage of by their own (most likely male) family members because of the economic incentive. Indeed, the project did not consider strategies that might encourage other members of the family to take over women’s responsibilities while they were participating in the workshop, nor did it discuss the use of the payment by the family members. Put differently, individuals who were encouraged to resist the imposition of climate change knowledge “from above” in workshops were submitted to potential oppression at home by the same intervention (Tschakert et al., 2016).

My examples from the field are not just about clumsy efforts to include women in workshops on climate change, or the tyranny of participation: These efforts still remain instruments of power to which policy-makers and project practitioners need to pay attention. While climate change knowledge politics are often discussed at the global level to highlight who gets to ask the questions that define global policies (e.g. Hulme, 2010), the local level and the knowledge *translation* and knowledge *collecting* part are often under-scrutinized.

4.4. Climate change adaptation politics

The politics of climate change adaptation is the fourth and final important analytical focus in this framework that, I argue, is useful for climate change policy makers and project practitioners who want to engage with the *workings of power*. To do so, adaptation politics need to be understood as an ensemble of policies, interventions, and everyday practices in the fashion of feminist political ecology, that is, one that revolves around the feminist concept of social reproduction. This understanding entails that the question that policy-makers and project practitioners need to ask is whether and how climate change politics can contribute to maintaining social reproduction—everyday life—and sustaining human cultures and communities on a daily basis and inter-generationally (Di Chico, 2008, 281). In Nicaragua, this understanding has clearly not been used; for example, climate change politics have helped to create a paternalistic relation between the government and rural women whose responsibilities were both naturalized and then increased, while at the same time women were expected to be grateful that they were the subjects of policies and measures that directly targeted them.

Nicaraguan climate change adaptation politics have not only been marked by the “feminization of responsibilities,” as this is also referred to in Chapter 13 of the IPCC’s fifth report (IPCC, 2014, 809), there has also been a paternalistic approach to women, incompatible with a just

and equitable vision of living life in common in the context of climate change. It is important to note that the paternalistic approach toward women is different from the paternalistic approach toward men in adaptation politics because it intervenes in a patriarchal context that is typical of many of the countries that make up the Global South wherein women—and especially those who are poor, indigenous, not supporting the government, and/or marginalized in other ways—are already facing increased oppression. The Nicaraguan government's relationship with rural women, as expressed through its policies, has been similar to the macho husband's relationship with his wife, or to the wealthy landowner's relations with the landless peasant workers on the hacienda (large capitalist farm). A similar system of gendered subordination has been put to work, but it has been hidden behind a gendered narrative and some gendered measures that are presented as just: the government provided women with 50 percent of the governmental positions, with pigs, chickens, plants, and micro-credit, which made the government “good” for women even when, in reality, it instrumentalized them and did not even consider challenging the interlinked causes and effects of patriarchy and climate change.

Hence, engaging with adaptation politics through the lens of social reproduction at all the interconnected scales can be useful for policy-makers and project practitioners. Through this lens, the main objectives of climate change politics become the sustainable reproduction of agricultural and societal systems that are socially and environmentally just and equitable now and in the future. This means that the objectives of such politics include challenging the interconnected causes and effects of patriarchy and climate change at the local and the global levels and also thinking about new ways of living life in common.

5. Tying it together: Policy-makers and project practitioners need to ask the right questions

Through the Nicaraguan example, I have argued that, in order to engage with the workings of power that make particular people vulnerable to climate change, policy-makers and practitioners need to pay attention to emotions, subjectivities, knowledges, and politics. First, emotions can be vehicles to transform existing power relations. Policies and projects should not manipulate emotions to ensure the maintenance of the status quo, and positive emotions need to be mobilized to help build alternative ways of living life in common in the context of climate change. Second, there is a need to engage with subjectivities in the climate change debate to raise crucial policy and project-relevant questions on the possibilities and pathways for emancipation from the status quo. Third, the politics of knowledge creation and knowledge translation on climate change adaptation should be given more

attention, in particular by addressing the ways in which participation may (re)produce marginalization on the ground. Finally, the politics of climate change adaptation need to be understood as an ensemble of policies, interventions, and everyday practices revolving around the feminist concept of social reproduction. This understanding has the advantage of including social reproductive chores such as firewood and water fetching, cooking, doing the laundry, and taking care of the children and the elderly, among environmental issues, without seeing them as immutable “traditional” roles of women and of opening up the space for talking about concepts such as care as part of adaptation.

Other scholars from the political ecology tradition have argued for drawing analytical attention to the four foci I have discussed above; they have done so by either engaging with them separately or by bringing some of them together (Tschakert et al., 2019, Nightingale, 2017, Manuel-Navarrete and Pelling, 2015, Tschakert et al., 2016, Eriksen et al., 2015). However, this work is rarely taken up by policy-makers and project practitioners because it seems to them too complex to operationalize. Table 1 presents some key questions policy-makers and project practitioners can ask to make sure that they engage with the workings of power in the processes that make particular people vulnerable to climate change and to open up the conceptual possibility of apprehending transformation and emancipation.

In cases such as the Nicaraguan one discussed above, I argue that asking questions like ones in Table 1, helps (re-)politicize the climate change adaptation debate and formulate climate change policies and projects that can become instruments of democratic governance. This is an urgent matter in contemporary Nicaragua where environmental issues (climate change among them) have been contributing to maintaining existing power hierarchies and indirectly leading to the population's insurrection that started in April 2018 in response to the consolidation of President Daniel Ortega's authoritarian populist political regime. Indeed, while I am writing these words in April 2019, for more than a year Nicaragua has been facing the worst political crisis it has experienced in recent decades. The Nicaraguan crisis has involved state-sponsored violations of human rights and violent repression that has resulted in hundreds of confirmed deaths, political prisoners, and the forced emigration of thousands of members of social, environmental, and peasant movements as well as the opposition in general. Nicaraguan climate change politics, in parallel with the manipulation of emotions, the creation of subjugated citizen-subjects, the reproduction of inequalities in knowledge-making processes along with the depletion of natural resources have contributed to consolidating the authoritarian regime of Daniel Ortega in the ways I have described in my vignettes from the field.

Table 1

Policy and project relevant questions for understanding the workings of power in the processes that make people vulnerable to climate change.

Main focus	Analytical foci	Questions for policy-makers and project practitioners
The <i>workings of power</i> in the processes that make people vulnerable to climate change	Emotions	How can we ensure that climate change policies and projects do not manipulate emotions? How can policies and projects build on positive emotions to create new pathways for transformation in the context of climate change?
	Subjectivities	How can the practices that support policies and projects' power to subjectivate their beneficiaries or participants be challenged? How can subjugated subjects become emancipated with the support of projects and policies? What do emancipatory subjectivities look like in the context of a changing environment and a society?
	Knowledges	How do existing power relations influence the translation of both “scientific” studies and input from local communities (especially from disadvantaged policy groups) into policies and project interventions? How can policy-makers and project practitioners creatively work with unequal power relations in climate change knowledge making and knowledge translating processes?
	Politics	What concrete policy and project measures are needed to envisage climate change adaptation as social reproduction and as a pathway for new ways of living life in common?

In general, there is a need to better understand and address the workings of the power-laden processes that make women and men vulnerable to climate change. The baseline is to stop seeing climate change adaptation as a “technical adjustment to biophysical change by society” and start considering it as “part of the dynamics of societies” (Eriksen et al., 2015, 2 italics in original) and “as both process and outcome” (Matyas and Pelling, 2015, 4 my italics) at the center of democratic governance debates. Political ecologists, human geographers, and development studies scholars need to better engage in discussions on climate change and gender not only to draw attention to the urgent necessity for radical transformations in how society deals with climate change but also to provide useful analytical tools that could feed into gender and climate change policy-making and project design. In the context of harshening climate change effects and the backlash in gender equality and democracy worldwide, there is, more than ever, an urgency to contribute to these efforts.

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